

World of Crayfish®

A short guide to completing the data template

Hello, and thank you for contributing to this project. This document explains each column of the template. The header uses Darwin Core standard terms where a clean equivalence exists, so that records stay interoperable with infrastructures such as GBIF; columns where the World of Crayfish® logic is specific keep their own meaning. The colours below match the colours of the template sections.

occurrenceID (*internally named WoCid*): the unique identifier of each record.

- Assigned automatically by the platform at upload. Contributors do not enter it, and it is not part of the template.
- Stable and unique: once assigned it does not change.
- It can be used as a repository accession number to cite or reference the record, including unpublished data deposited in World of Crayfish®.

Associated taxa require a host. Every record is a crayfish occurrence (section 3). The associated-taxa section (section 4) is optional and is completed only when a pathogen, symbiont, or other organism is found on that crayfish. We do not accept associated taxa without a crayfish host.

1) Source of the information

bibliographicCitation (*a.k.a. DOI*)

- The unique DOI of the source article. Mandatory if a DOI exists; otherwise leave blank.
- Caution: when dragging to auto-fill in Excel, the DOI may be altered into successive increasing values. Check after filling.

associatedReferences (*a.k.a. URL*)

- The URL where the article can be found. Required only if no DOI is available.

sourceCitation (*a.k.a. full citation*)

- The citation text in APA format (find the title in Google Scholar, click the cite symbol, select APA).
- Mandatory if no DOI is available; otherwise not required.
- If the data are unpublished but yours, or you have permission to use them, type unpublished.

2) Location and accuracy

decimalLatitude / decimalLongitude

- Mandatory. Each coordinate in its own column; do not mix latitude and longitude.

- Decimal Degrees recommended, numeric only, no symbols (°, ', ", N, S, E, W). Use a minus sign for the western hemisphere (longitude) and the southern hemisphere (latitude). Example: Lat 24.1894, Long 14.3236.
- DMS and DDM formats are accepted but converted internally.

accuracy

- A flag for how accurate the coordinates are relative to the field observation. Mandatory.

Value	Meaning	Precision
high	Extracted from published maps at hydrographic level; from detailed published toponymy; or exact coordinates.	Pinpoints the precise river or lake; useful for SDM.
low	From continental or national grid systems; estimated from national-level maps; from low-detail toponymy.	Cannot pinpoint the precise river or lake; useful for range estimation.

Note. Do not store multiple records in the same spot for the same species and the same year. Use the extinction function (below) to record disappearance of a previously recorded population. The same applies to *A. astaci*: record extinction when an infection outbreak was certainly documented as eradicated.

3) Crayfish species observed

scientificName

- The scientific name of the observed crayfish species. Mandatory.

establishmentMeans (a.k.a. population status)

- Characterises the status of the meta-population. Mandatory.
- Accepted values: indigenous or non-indigenous .
- Kept distinct from occurrenceOrigin , and it constrains which origins are allowed (see below).

occurrenceOrigin

- Characterises the provenance of the occurrence. Mandatory. This is a World of Crayfish® field, kept as is; it is not merged into establishmentMeans .
- The allowed values **depend on** establishmentMeans :

If establishmentMeans is	occurrenceOrigin can be
indigenous	native, type locality
non-indigenous	introduced, invasive, cryptogenic

- Use type locality when indexing the paper in which a species was described; this displays the type locality on the map for future searches.

year

- The year of the observation (or of the documented extinction). Mandatory; if unknown, use the year of publication.

associatedSequences (a.k.a. NCBI / GenBank accession code)

- The GenBank accession number for any sequence type (for example COI, 16S, mitogenome), e.g. `MK645042`, with no sequence-type label. Optional; multiple codes separated by commas.

catalogNumber

- The identifier of a voucher specimen for this occurrence, if a physical specimen exists (for example a museum catalogue number). Optional; leave blank for observation-only records.

institutionCode

- The institution or collection holding the voucher specimen (for example USNM, MNHN). Optional; complete it when catalogNumber is filled.

Records with a voucher specimen (catalogNumber filled) are flagged as preserved specimens at export, so end users can filter for material available for study.

claimExtinction

- If there is certain information that a crayfish population has completely disappeared from a locality, add a distinct record and type `Extinct` here. This flags the locality as extirpated on current distribution maps, starting from the year in `year`.
- Local extirpation defaults to a 500 m radius buffer around the location; set your own value in `extirpationBuffer` (metres).
- This must be a distinct line, with an appropriate source (including `Unpublished` if assumed by an expert) proving the extinction.
- Use `occurrenceRemarks` to describe the cause, using: `invasive: [species]`, `disease: [pathogen]`, `habitat loss: [description]`, `combined invasive + disease`, or `unknown`.

extirpationBuffer

- The custom local-extirpation radius in metres. Optional; default 500.

4) Associated taxa (pathogens, symbionts, epibionts)

associatedTaxa

- The scientific name of an organism associated with the crayfish: a pathogen, symbiont, epibiont, or other associate (for example *Aphanomyces astaci*, *Branchiobdella* spp., a virus). Optional.
- Always recorded on a crayfish host named in section 3. We do not accept associated taxa without a host.

associatedTaxaSequences

- The GenBank accession for the associated taxon (not the crayfish host). Optional.

5) Additional information

occurrenceRemarks *(a.k.a. comments)*

- Any additional information worth mentioning. Optional. Also used to describe the cause of a local extinction (see `claimExtinction`).

confidentialityLevel

- The degree of openness chosen for the data.
- `0` or blank: no restrictions; data shown and downloaded with exact location.
- `1`: normal restrictions; non-registered users see a 25 km hexagon, registered users see the exact location.
- `2`: high restrictions; all users see a 25 km hexagon, registered users see the exact location only on a well-reasoned request approved by the IAA.

contributor

- The name of the contributor to the database (not necessarily the author of the source paper). Mandatory.

Two axes are deliberately separate. `establishmentMeans` (meta-population status) and `occurrenceOrigin` (record provenance) are distinct fields and are never merged. Both must be completed, and the chosen status constrains the allowed origin.

Save the final document with the label **WoC-Country-Name**.

Important. World of Crayfish® can be a repository for your distribution datasets. We can provide accession codes for your unpublished records.